

Truncated Microlocal Morse Theory

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1 Introduction

Microlocal Morse theory is machinery to study the propagation of cohomology of complexes of sheaves. More generally, this theory is considered as a part of microlocal sheaf theory, due to Kashiwara and Schapira. They defined the notion of microsupport of a complex of sheaves in the 1980's.

To explain more precisely, let us fix some notations. We consider a commutative unital ring $\mathbf{k}$ of finite global dimension (e.g., $\mathbf{Z}$ or a field). Let X be a topological space. We denote by $D^b(\mathbf{k}_X)$ (resp. $D^+(\mathbf{k}_X)$) the bounded derived category (resp. bounded derived category from below) of sheaves of $\mathbf{k}$ -modules on X .

If M is a real manifold of class C^∞ and F a complex of sheaves, then the microsupport $SS(F)$ of F is a closed conic (i.e., stable under $\mathbf{R}^+$ -action) involutive (i.e., coisotropic) subset of the cotangent bundle of M . The precise definition is given in Section 3, but roughly speaking, $SS(F)$ describes the (co)directions of X in which the cohomology of F cannot be extended. This notion can be interpreted by analogy with Morse theory. Let φ be a real-valued function of class C^1 . Then we can replace the notion of critical points of φ with the one with respect to F . That is to say, we can define a critical point with respect to F by the point p such that $d\varphi(p)$ belongs to $SS(F)$. We can then obtain an analogous result of Morse theory in a framework of sheaf theory. In Section 5, we prove a truncated version of (a particular case of) the microlocal Morse lemma (cf. [GKS12, Proposition 1.8] or [KS90, Corollary 5.4.19]). Roughly speaking, this asserts that if there is no critical point with respect to a complex of sheaves, we have the natural isomorphism of the cohomologies of the sublevel sets of the manifold.

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In 2003, Kashiwara, Monteiro-Fernandes, and Schapira introduced the notion of truncated microsupport in [KMS03a], in the course of the study of solutions of boundary value problems of $\mathcal{D}$ -modules. The difference between microsupports and truncated microsupports is as follows. On the one hand, a point p in the cotangent bundle T^*M does not belong to the microsupport if and only if a certain kind of cohomologies of all the degrees vanish. On the other hand, the point p does not belong to the truncated microsupport if and only if the same kind of cohomologies of all the degrees less than a fixed number vanish.

After they introduced truncated microsupports, some properties of truncated microsupports were shown. For example, a truncated microsupport is not involutive, but has a property called weak involutivity. This is shown also by Kashiwara, Monteiro-Fernandes, and Schapira in [KMS03b]. Some functorial properties of truncated microsupport are studied by several authors. See [KMS03a, FM07]. These authors studied truncated microsupports mainly in the context of analytic problems.

In this paper, in contrast to the previous research, we shall apply the truncated microsupport to the study of some geometrical problems. First we prove a truncated version of the noncharacteristic deformation lemma. Consider a Hausdorff space and a family of open subsets of it. Assume that the family is “noncharacteristic” with respect to a complex of sheaves on the space. In this case, the original version of the theorem asserts that the global section on a member of the family is isomorphic to the one on the union of the family. Note that this isomorphism is one between derived functors.

This lemma is one of the most fundamental results in microlocal sheaf theory, and used throughout [KS90]. Thus a truncated version of this lemma is also the key result in this paper. However, one should notice that the original proof of the lemma cannot be applied to our truncated version. To prove the result, we make three modifications to the lemma. First, we add an assumption on a total space. That is, we assume that the space is not only Hausdorff, but also satisfies the condition that all the open subspaces are paracompact. Second, we weaken the hypothesis of vanishing of cohomologies. In the original lemma, we assume that the cohomologies of all degrees vanish. However, in our version, we assume that the cohomologies of not all degrees but degrees less than a certain number vanish. Third, we make an additional assumption on the family of deformation. Therefore, to make, in particular, the third assumption acceptable to the reader, we give several examples of families of deformation with figures. Then, in turn, we formulate a truncated version (Theorem 5.3) of the microlocal Morse lemma ([KS90,

Proposition 5.4.17]), under the light of truncated microsupport. Note that we can also study the unique continuation property of cohomology groups of sheaves through the truncated microsupport. More precisely, if k is an integer and we consider the truncated microsupport $\mathrm{SS}_k(F)$ of a bounded complex of sheaves F , we can investigate whether the cohomology sheaves $H^j(F)$ propagate or not for $j < k$. Furthermore, we can see whether the cohomology sheaves $H^k(F)$ of the k -th degree enjoy the unique continuation property or not.

The paper is organized as follows. In Section 2, we review the notations and results from [KS90]. In Section 3, we define truncated microsupport of sheaves, due to [KMS03a]. In Section 4, we prove a truncated version of the noncharacteristic deformation lemma. In the proof, Lemma 4.4 plays a crucial role. However, since the proof of the lemma is long and rather technical, its proof will be given in Section 6. Finally in Section 5, we apply the result in Section 4 to the microlocal Morse theory.

2 Background Material

In this section, we recall some definitions and results of homological algebra and sheaf theory including the theory of derived categories from [KS90], following its notations with slight modifications.

Let X be a topological space, and Z a locally closed set of X . Let us denote by $i: Z \hookrightarrow X$ the inclusion. We use the following functors:

$$\mathrm{R}\Gamma_Z(\cdot) := \mathrm{R}i_*i^!, \quad (\cdot)_Z := \mathrm{R}i_!i^{-1}.$$

If there is no risk of confusion, we will write $\mathbf{k}_Z$ rather than $\mathbf{k}_{X,Z}$.

2.1 Geometrical Notions

2.1.1 Proper Morphisms

We use the following proposition in the proof of Proposition 5.1

Proposition 2.1 (see [Wed16, Proposition 1.32]). Let X , Y , and Z be locally compact (Hausdorff, by definition) spaces, and let f, g be continuous. If $g \circ f$ is proper and f surjective, then g is proper.

2.1.2 The Fundamental Diagram of (co)tangent bundles

Let M be a real manifold of class C^∞ . We denote by $\tau_M: TM \rightarrow M$ the tangent bundle of M , and by $\pi_M: T^*M \rightarrow M$ the cotangent bundle of M . If

there is no risk of confusion, we will write π rather than π_M . If $f: M \rightarrow N$ is a morphism of manifolds (i.e., a C^∞ -map), then we have the following natural diagrams.

$$\begin{array}{ccccc}
TM & \xrightarrow{f'} & M \times_N TN & \xrightarrow{f_\tau} & TN \\
& \searrow \tau_M & \downarrow & \square & \downarrow \tau_N \\
& & M & \xrightarrow{f} & N,
\end{array}
\qquad
\begin{array}{ccccc}
T^*M & \xleftarrow{f_d} & M \times_N T^*N & \xrightarrow{f_\pi} & T^*N \\
& \searrow \pi_M & \downarrow & \square & \downarrow \pi_N \\
& & M & \xrightarrow{f} & N.
\end{array}$$

In these diagrams, f_τ and f_π denote the natural projections, f' is defined by $(x; v) \mapsto (x; df_x v)$, and f_d is induced by the transposition of f' .

2.2 Approximation by Inductive Limits

Proposition 2.2 ([KS90, Proposition 2.5.1]). Let X be a topological space, Z a subspace, and F a sheaf on X . Consider the canonical morphism:

$$\psi: \varinjlim_U \Gamma(U; F) \rightarrow \Gamma(Z; F),$$

where U ranges through the family of open neighborhoods of Z in X . Then the following hold.

- (i) The morphism ψ is injective.
- (ii) Assume that X is Hausdorff and Z is compact. Then ψ is an isomorphism.
- (iii) Assume that X is paracompact and Z is closed. Then ψ is an isomorphism.

Note that a paracompact space is, by definition, a Hausdorff space.

2.3 Complement of Homological Algebra

Consider a projective system of abelian groups $(M_n, \rho_{n,p})_n$ indexed by $\mathbf{N}$. One says that this system satisfies the Mittag-Leffler condition (ML condition for short) if for any $n \in \mathbf{N}$, the decreasing sequence $(\rho_{n,p}(M_p))_p$ of subgroups of M_n is stationary. For a projective system of abelian groups $(M_n)_n$, we set

$$M_\infty := \varprojlim_n M_n.$$

Consider a projective system of complexes

$$(2.1) \quad E_n^\bullet: \cdots \rightarrow M_n^{j-1} \rightarrow M_n^j \rightarrow M_n^{j+1} \rightarrow \cdots,$$

and its projective limit

$$(2.2) \quad E_\infty^\bullet: \cdots \rightarrow M_\infty^{j-1} \rightarrow M_\infty^j \rightarrow M_\infty^{j+1} \rightarrow \cdots.$$

Denote by

$$(2.3) \quad \Phi_k: H^k(E_\infty^\bullet) \rightarrow \varprojlim_n H^k(E_n^\bullet)$$

the natural morphism.

Proposition 2.3 ([KS90, Proposition 1.12.4]). Assume that for each $k \in \mathbf{Z}$, the system $(M_n^k)_n$ satisfies the ML condition. Then

- (a) for each $k \in \mathbf{Z}$, the map Φ_k is surjective,
- (b) if moreover, for a given i the system $(H^{i-1}(E_n^\bullet))_n$ satisfies the ML condition, then Φ_i is bijective.

The following is called Kashiwara's lemma.

Proposition 2.4 ([KS90, Proposition 1.12.6]). Let $(X_s, \rho_{s,t})$ be a projective system of sets indexed by $\mathbf{R}$. Assume that for each $s \in \mathbf{R}$, the canonical maps

$$\lambda_s: X_s \rightarrow \varprojlim_{r < s} X_r$$

and

$$\mu_s: \varinjlim_{t > s} X_t \rightarrow X_s$$

are both injective (resp. surjective). Then all the maps ρ_{s_0, s_1} ($s_0 \leq s_1$) are injective (resp. surjective).

2.4 ML Procedure for Sheaves

We use the following in the proof of Theorem 4.2.

Proposition 2.5 ([KS90, Proposition 2.7.1]). Let X be a topological space and let $F \in \mathbf{D}^+(\mathbf{k}_X)$. Let $(U_n)_{n \in \mathbf{N}}$ be an increasing sequence of open subsets of X , $(Z_n)_{n \in \mathbf{N}}$ a decreasing sequence of closed subsets of X . Set $U := \bigcup_n U_n$, $Z := \bigcap_n Z_n$. Then the following hold.

- (i) For any j , the natural map $\varphi_j: H_Z^j(U; F) \rightarrow \varprojlim_n H_{Z_n}^j(U_n; F)$ is surjective.
- (ii) Assume that for a given j , the projective system $(H_{Z_n}^{j-1}(U_n; F))_n$ satisfies the ML condition. Then φ_j is bijective.
- (iii) Let now $(X_n)_{n \in \mathbf{N}}$ be an increasing family of subsets of X satisfying $X = \bigcup_n X_n$ and $X_n \subset \text{Int}(X_{n+1})$ for all n . Assume that for a given j , the projective system $(H^{j-1}(X_n; F))_n$ satisfies the ML condition. Then the natural map $H^j(X; F) \rightarrow \varprojlim_n H^j(X_n; F)$ is bijective.

3 Truncated Microsupport

In this section, we review the definition of a truncated microsupport and its functorial property under direct images of proper morphisms. First, we review the original definition of a microsupport.

Definition 3.1 ([KS90, Definition 5.1.2]). Let $F \in \mathbf{D}^b(\mathbf{k}_X)$. The closed conic subset $\text{SS}(F)$ of T^*X is defined by the following condition: $p \notin \text{SS}(F)$ if and only if there exists an open neighborhood U of p such that for any $x \in \pi(U)$ and for any real-valued C^1 -function φ defined on a neighborhood of x such that $\varphi(x) = 0$, $d\varphi(x) \in U$, one has

$$(3.1) \quad \text{R}\Gamma_{\{\varphi \geq 0\}}(F)_x = 0.$$

Kashiwara, Monteiro-Fernandes, and Schapira modified this definition and gave the notion of truncated microsupport.

Definition 3.2 ([KMS03a, Definition 3.4]). Let $F \in \mathbf{D}^b(\mathbf{k}_X)$, and fix an integer $k \in \mathbf{Z}$. The *truncated microsupport* $\text{SS}_k(F)$ is the closed conic subset of T^*X defined by the following condition: $p \notin \text{SS}_k(F)$ if and only if there exists an open conic neighborhood U of p such that for any $x \in \pi(U)$ and for any real-valued C^1 -function φ defined on a neighborhood of x such that $\varphi(x) = 0$, $d\varphi(x) \in U$, one has

$$(3.2) \quad H_{\{\varphi \geq 0\}}^j(F)_x = 0 \quad \text{for any } j \leq k.$$

For other equivalent conditions of the definition, see [KMS03a]. Truncated microsupports satisfy the following functorial property under proper direct images.

Proposition 3.3. Let $f: X \rightarrow Y$ be a morphism of manifolds and let $F \in \mathbf{D}^b(\mathbf{k}_X)$ such that f is proper on $\text{supp}(F)$. Then for any $k \in \mathbf{Z}$,

$$(3.3) \quad \text{SS}_k(\mathbf{R}f_*F) \subset f_\pi f_d^{-1}(\text{SS}_k(F)).$$

Moreover if f is a closed embedding, the inclusion above is an equality.

For other functorial properties, see [KMS03a, FM07].

4 Truncated Noncharacteristic Deformation Lemma

In this section, we prove a truncated version of the noncharacteristic deformation lemma. First we state the result. After giving several examples, we prove Theorem 4.2. In the course of the proof, we use the key lemma (Lemma 4.4). Since its proof is rather long, we postpone the proof to Section 6.

Before we state the truncated version, let us review the original version of the noncharacteristic deformation lemma.

Proposition 4.1 ([KS90, Proposition 2.7.2]). Let X be a Hausdorff space. Let $F \in \mathbf{D}^+(\mathbf{k}_X)$, and let $(U_t)_{t \in \mathbf{R}}$ be a family of open subsets of X . We assume the following conditions.

- (i) $U_t = \bigcup_{s < t} U_s$, for all $t \in \mathbf{R}$.
- (ii) For all pairs (s, t) with $s \leq t$, the set $\overline{U_t \setminus U_s} \cap \text{supp } F$ is compact.
- (iii) Setting $Z_s := \bigcap_{t > s} \overline{U_t \setminus U_s}$, we have for all pairs (s, t) with $s \leq t$ and all $x \in Z_s \setminus U_t$,
$$(\mathbf{R}\Gamma_{X \setminus U_t}(F))_x = 0.$$

Then, for all $t \in \mathbf{R}$, we have the isomorphisms

$$\mathbf{R}\Gamma\left(\bigcup_{s \in \mathbf{R}} U_s; F\right) \xrightarrow{\sim} \mathbf{R}\Gamma(U_t; F).$$

Our result is as follows.

Theorem 4.2 (Truncated Noncharacteristic Deformation Lemma). Let X be a Hausdorff space. Let $F \in \mathbf{D}^+(\mathbf{k}_X)$, and let $(U_t)_{t \in \mathbf{R}}$ be a family of open subsets of X . Let $k \in \mathbf{Z}$. For $s \in \mathbf{R}$, set $Z_s := \bigcap_{t > s} \overline{U_t \setminus U_s}$. We assume the following conditions.

- (i) $U_t = \bigcup_{s < t} U_s$, for all $t \in \mathbf{R}$.
- (ii) For all pairs (s, t) with $s \leq t$, the set $\overline{U_t \setminus U_s} \cap \text{supp } F$ is compact.
- (iii) For all $s \in \mathbf{R}$ and all $x \in Z_s \setminus U_s$, we have

$$(H_{X \setminus U_s}^j(F))_x = 0 \quad \text{for all } j \leq k.$$

- (iv) For all pairs (s, t) with $s < t$, we have $Z_s \subset U_t$.

Then, for all $t \in \mathbf{R}$, we have the isomorphisms

$$H^j \left(\bigcup_{s \in \mathbf{R}} U_s; F \right) \xrightarrow{\sim} H^j(U_t; F) \quad \text{for all } j < k,$$

and the monomorphisms

$$H^k \left(\bigcup_{s \in \mathbf{R}} U_s; F \right) \hookrightarrow H^k(U_t; F).$$

Before we prove Theorem 4.2, let us give several examples of families of deformation.

Example 4.3. The condition (iv) of Theorem 4.2 is not contained in the original version (Proposition 4.1). Therefore let us give an example of $(U_t)_{t \in \mathbf{R}}$ that does not satisfy the condition (iv).

- (i) Let $X = \mathbf{R}^2$, and for a real number t define the function

$$\varphi_t(x) := \begin{cases} \max(0, t(2 - |x|)) & (|x| < 2), \\ 0 & (|x| \geq 2). \end{cases}$$

Then let us define the open subset U_t of X by

$$U_t := \{(x, y) \in X ; y < \varphi_t(x)\}.$$

Then the family $(U_t)_t$ does not satisfy the condition (iv). Indeed, if $0 < s < t$, we have

$$Z_s = \{(x, y) \in X; y = \varphi_s(x), |x| \leq 2\}$$

and the point $(2, 0) \in Z_s$ is not contained in U_t . Moreover, let us define a locally closed subset S of X by

$$S := \{(x, y) \in X; (x + 2)^2 + (y - 1)^2 = 1\} \setminus \{(-1, 0)\}.$$

We denote by $\mathbf{k}_S$ the sheaf on X as in Section 2. If $j \leq 0$, we have $H_{X \setminus U_t}^j(\mathbf{k}_S)_x = 0$ for all $t \geq s$ and $x \in Z_s \setminus U_t$, but the morphism $H^0(U_1; \mathbf{k}_S) = \mathbf{k} \rightarrow 0 = H^0(U_0; \mathbf{k}_S)$ is not injective. This gives an example that satisfies the conditions (i)–(iii) but not (iv), which results in the consequence of the proposition does not hold. Hence the condition (iv) is necessary.

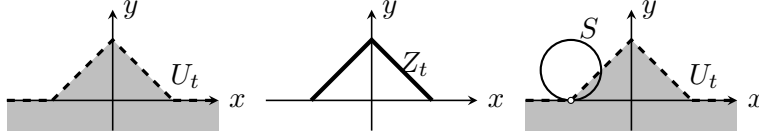


Figure 1: example (i)

Let us, in turn, give two examples of $(U_t)_{t \in \mathbf{R}}$ satisfying the condition (iv). In all the examples below, X is $\mathbf{R}^2$.

(ii) For a real number t , define the open set U_t of X by

$$U_t := \begin{cases} \{(x, y) \in X; x^2 + y^2 < t\} & (t > 0), \\ \emptyset & (t \leq 0). \end{cases}$$

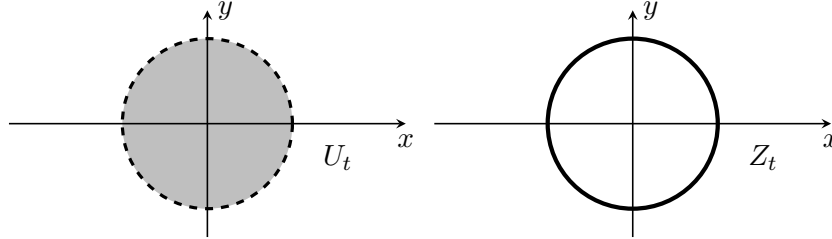


Figure 2: example (ii)

(iii) For a real number t , define the function $\varphi_t: \mathbf{R} \rightarrow \mathbf{R}$ as follows. If $t \leq 0$, set $\varphi_t(x) \equiv 0$. If $t > 0$, set

$$\varphi_t(x) := \begin{cases} \sqrt{t^2 - x^2} & (|x| < t), \\ 0 & (|x| \geq t). \end{cases}$$

Then define the open set U_t of X by

$$U_t := \{(x, y) \in X; y < \varphi_t(x)\}.$$

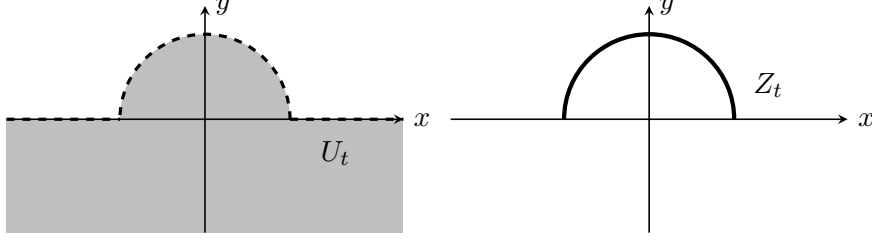


Figure 3: example (iii)

Proof of Theorem 4.2 Consider the following conditions for $j \in \mathbf{Z}$ and $s \in \mathbf{R}$:

$$\begin{aligned}
 (a)_s^j & \quad \mu_s^j: \varinjlim_{t>s} H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F), \\
 (a')_s^k & \quad \mu_s^k: \varinjlim_{t>s} H^k(U_t; F) \hookrightarrow H^k(U_s; F), \\
 (b)_s^j & \quad \lambda_s^j: H^j(U_s; F) \xrightarrow{\sim} \varprojlim_{r<s} H^j(U_r; F).
 \end{aligned}$$

Let us prove, for all $s \in \mathbf{R}$, $(a)_s^j$ for $j < k$ and $(a')_s^k$. We may assume that $\overline{U_t \setminus U_s}$ is compact for any $s \leq t$ by replacing X by $\text{supp } F$. Consider the distinguished triangle

$$\text{R}\Gamma_{X \setminus U_s}(F) \rightarrow F \rightarrow \text{R}\Gamma_{U_s}(F) \xrightarrow{+1}.$$

Let us endow $Z_s \cup U_s$ with the induced topology, and denote by $i_s: Z_s \cup U_s \hookrightarrow X$ the natural continuous map. Applying the functor

$$(\cdot)|_{Z_s \cup U_s} = i_s^{-1}$$

to the distinguished triangle above, we have

$$\text{R}\Gamma_{X \setminus U_s}(F)|_{Z_s \cup U_s} \rightarrow F|_{Z_s \cup U_s} \rightarrow \text{R}\Gamma_{U_s}(F)|_{Z_s \cup U_s} \xrightarrow{+1}.$$

Again applying to this triangle the functor $\text{R}\Gamma(Z_s \cup U_s; \cdot)$, we have

$$(4.1) \quad \text{R}\Gamma(Z_s \cup U_s; \text{R}\Gamma_{X \setminus U_s}(F)|_{Z_s \cup U_s}) \rightarrow \text{R}\Gamma(Z_s \cup U_s; F|_{Z_s \cup U_s}) \rightarrow \text{R}\Gamma(U_s; F) \xrightarrow{+1}.$$

By the hypothesis (iii) and the definition of a local cohomology, we see that

$$H_{X \setminus U_s}^j(F)|_{Z_s \cup U_s} = 0 \quad \text{for any } j \leq k.$$

Therefore we have

$$H^j(Z_s \cup U_s; \mathbf{R}\Gamma_{X \setminus U_s}(F)|_{Z_s \cup U_s}) = 0 \quad \text{for any } j \leq k.$$

Thus we have from (4.1)

$$(4.2) \quad H^j(Z_s \cup U_s; F|_{Z_s \cup U_s}) \xrightarrow{\sim} H^j(U_s; F) \quad \text{for any } j < k,$$

$$(4.3) \quad H^k(Z_s \cup U_s; F|_{Z_s \cup U_s}) \hookrightarrow H^k(U_s; F).$$

Therefore, if we show Lemma 4.4 below, whose proof will be given in Section 6, we have from Lemma 4.4 and (4.2)

$$\begin{aligned} \varinjlim_{t>s} H^j(U_t; F) &\cong H^j(Z_s \cup U_s; F|_{Z_s \cup U_s}) \\ &\xrightarrow{\sim} H^j(U_s; F) \end{aligned}$$

for any $j < k$, and $(a)_s^j$ follows. Similarly, we have from Lemma 4.4 and (4.3)

$$\begin{aligned} \varinjlim_{t>s} H^k(U_t; F) &\cong H^k(Z_s \cup U_s; F|_{Z_s \cup U_s}) \\ &\hookrightarrow H^k(U_s; F) \end{aligned}$$

and $(a')_s^k$ follows.

Lemma 4.4. For any $j \in \mathbf{Z}$ and $s \in \mathbf{R}$, there exists a natural isomorphism

$$\varinjlim_{t>s} H^j(U_t; F) \cong H^j(Z_s \cup U_s; F|_{Z_s \cup U_s}).$$

We prove $(b)_s^j$ for all $j \in \mathbf{Z}$ and $s \in \mathbf{R}$ by induction on j . Since $F \in \mathbf{D}^+(\mathbf{k}_X)$, we have $(b)_s^j$ for $j \ll 0$ and $s \in \mathbf{R}$. Hence there exists an integer $j_0 < k$ such that $(b)_s^j$ holds for any $j < j_0$ and $s \in \mathbf{R}$. Then assume $(b)_s^j$ is proved for all $j < j_0$ and $s \in \mathbf{R}$. Since we have $(a)_s^j$ and $(b)_s^j$ for each $j < j_0$ and each $s \in \mathbf{R}$, we can apply Kashiwara's lemma (Proposition 2.4) and have

$$H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F) \quad \text{for } j < j_0 \text{ and } s \leq t.$$

Fix $s \in \mathbf{R}$ and consider the projective system $(H^{j_0-1}(U_{s-1/n}); F)_{n \in \mathbf{N}}$. This system satisfies the ML condition. Therefore, by applying Proposition 2.5 (iii), we have

$$H^j(U_s; F) \xrightarrow{\sim} \varprojlim_n H^j\left(U_{s-\frac{1}{n}}; F\right).$$

Since $(s-1/n)_n$ is a cofinal family of $(t)_{s>t}$, we have $(b)_s^{j_0}$. Then by induction on j , we have $(b)_s^j$ for $j < k$ and $s \in \mathbf{R}$. Again by applying Proposition 2.5 to the projective system $(H^j(U_n; F))_n$, we obtain

$$\begin{aligned} H^j \left(\bigcup_{s \in \mathbf{R}} U_s; F \right) &\cong H^j \left(\bigcup_{n \in \mathbf{N}} U_n; F \right) \\ &\xrightarrow[\leftarrow]{n} \varprojlim_n H^j(U_n; F) \end{aligned}$$

and for each n , we have by Kashiwara's lemma (Proposition 2.4) that

$$H^j(U_n; F) \cong H^j(U_s; F)$$

for $n \geq s$.

Finally we replace $(a)_s^j$ by $(a')_s^k$ for $j = k$, and obtain

$$H^k \left(\bigcup_{s \in \mathbf{R}} U_s; F \right) \hookrightarrow H^k(U_s; F).$$

This completes the proof. $\square$

5 Truncated microlocal Morse theory

Let $f: Y \rightarrow X$ be a morphism of manifolds, and φ a real valued function of class C^1 . For a real number $t \in \mathbf{R}$, we set

$$\begin{aligned} Y_t &:= \{y \in Y ; \varphi(y) < t\} = \varphi^{-1}(]-\infty, t[), \\ \tilde{Y}_t &:= \{y \in Y ; \varphi(y) \leq t\} = \varphi^{-1}(]-\infty, t]). \end{aligned}$$

Let us define the following morphisms:

$$\begin{array}{ccc} Y_t & \xrightarrow{j_t} & Y \\ & \searrow f_t & \downarrow f \\ & & X, \end{array} \quad \begin{array}{ccc} \tilde{Y}_t & \xrightarrow{\tilde{j}_t} & Y \\ & \searrow f_t & \downarrow f \\ & & X. \end{array}$$

Proposition 5.1. Let $G \in D^b(Y)$, and assume $f_t: \text{supp}(G) \cap \overline{Y}_t \rightarrow X$ is proper for all $t \in \mathbf{R}$. Let k be an integer and t_0 a real number. Assume that, for any $y \in Y \setminus \tilde{Y}_{t_0}$, we have

$$(5.1) \quad d\varphi(y) \notin \left(\text{SS}_k(G) + f_d \left(Y \times_X T^*X \right) \right).$$

Then the following statements hold.

(a) For $t > t_o$, the natural morphism

$$R^j f_* G \rightarrow R^j f_{t*} j_t^{-1} G$$

is an isomorphism for $j < k$ and injective for $j = k$.

(b) We have

$$\begin{aligned} \mathrm{SS}_k(Rf_* G) &\subset f_\pi \left(f_d^{-1} \left(\mathrm{SS}_k(G) \cap \pi^{-1}(\tilde{Y}_{t_o}) \right) \right) \\ &= f_\pi f_d^{-1}(\mathrm{SS}_k(G)). \end{aligned}$$

Proof. Following [KS90], let $\tilde{f}$ be the morphism $(f, \varphi): Y \rightarrow X \times \mathbf{R}$, and q and $\tilde{\varphi}$ be the first and second projections defined on $X \times \mathbf{R}$ respectively. Then $\tilde{f}$ is proper on $\mathrm{supp}(G)$ and $f = q \circ \tilde{f}$. Let i_t be the open embedding $X \times]-\infty, t[\hookrightarrow X \times \mathbf{R}$ and $\tilde{i}_t$ the closed embedding $X \times]-\infty, t] \hookrightarrow X \times \mathbf{R}$. Set $q_t = q \circ i_t$ and $\tilde{q}_t = q \circ \tilde{i}_t$. These morphisms can be visualized by the following diagrams, in both of which the left squares are cartesian.

$$\begin{array}{ccc} Y & \xrightarrow{\tilde{f}} & X \times \mathbf{R} \xrightarrow{q} X, \\ j_t \uparrow & \square & \uparrow i_t \\ Y_t & \xrightarrow{\tilde{f}_t} & X \times]-\infty, t[\end{array} \quad , \quad \begin{array}{ccc} Y & \xrightarrow{\tilde{f}} & X \times \mathbf{R} \xrightarrow{q} X. \\ \tilde{j}_t \uparrow & \square & \uparrow \tilde{i}_t \\ Y_t & \xrightarrow{\tilde{f}_t} & X \times]-\infty, t] \end{array}$$

Then, setting $\tilde{G} := R\tilde{f}_* G$, we have

$$\begin{aligned} Rf_* G &\cong Rq_* \tilde{G}, & R\tilde{f}_* G &\cong R\tilde{q}_* \tilde{G}, \\ Rf_{t*} j_t^{-1} G &\cong Rq_{t*} i_t^{-1} \tilde{G}, & R\tilde{f}_{t*} \tilde{j}_t^{-1} G &\cong R\tilde{q}_{t*} \tilde{i}_t^{-1} \tilde{G}, \\ Rf_{t!} j_t^{-1} G &\cong Rq_{t!} i_t^{-1} \tilde{G}, & R\tilde{f}_{t*} \tilde{j}_t^! G &\cong R\tilde{q}_{t*} \tilde{i}_t^! \tilde{G}. \end{aligned}$$

Therefore, it is enough to prove the result for $\tilde{G}$, q , i_t , $\tilde{\varphi}$ instead of G , f , j_t , φ .

(i) It follows from the hypothesis (5.1) that

$$(5.2) \quad \mathrm{SS}_k(\tilde{G}) \cap (T^*X \times \{(t; \tau); t > t_o\}) \subset T^*X \times \{(t; \tau); \tau \leq 0\}.$$

Indeed, by hypothesis, for any $(y; \eta) \in \mathrm{SS}_k(G)$ and $(f(x); \xi) \in Y \times_X T^*X$, we have

$$(5.3) \quad d\varphi(y) \neq \eta + f^* \xi.$$

If $(x, t; \xi, \tau)$ is a point of $\text{SS}_k(\tilde{G}) \cap (T^*X \times \{(t; \tau); t > t_0\})$, then, by the formula (3.3), there exists $(y; \eta) \in \text{SS}_k(G)$ such that

$$\tilde{f}(y) = (x, t) \text{ and } \eta = \tilde{f}^*(\xi, \tau).$$

For this η , we have

$$\eta = \tilde{f}^*(\xi, \tau) = f^*\xi + \varphi^*\tau = f^*\xi + \tau d\varphi(y).$$

If $\tau > 0$, we get

$$d\varphi(y) = \frac{1}{\tau}\eta - \frac{1}{\tau}f^*\xi$$

and this contradicts (5.3) because $\text{SS}_k(G)$ is conic.

We shall show that for all $x \in X$

$$\text{R}^j q_* \tilde{G}_x \rightarrow \text{R}^j q_{t*} i_t^{-1} \tilde{G}_x$$

is an isomorphism if $j < k$ and injective if $j = k$. By definition, it suffices to show the statement for the morphism

$$\varinjlim_{x \in U} H^j(U \times \mathbf{R}; \tilde{G}) \rightarrow \varinjlim_{x \in U} H^j(U \times]-\infty, t[; \tilde{G}).$$

Let U be a neighborhood of x , open in X contained in a local chart, and diffeomorphic to the unit open ball, centered at 0. Such open sets form a fundamental system of neighborhoods of x . Thus we show the result for the morphism

$$H^j(U \times \mathbf{R}; \tilde{G}) \rightarrow H^j(U \times]-\infty, t[; \tilde{G}).$$

when U is in the form as above.

We shall construct a deformation family $(U_s)_{s \in \mathbf{R}}$ and apply Theorem 4.2 to it. First, Define a function $\chi_s: U \rightarrow \mathbf{R}$ by

$$\chi_s(x) := \begin{cases} t & (s \leq t), \\ -4|x| + 2(s - t) & (t < s \leq t + 1), \\ -\frac{s-2}{1-1/2s}|x| + (s - 2) & (s \geq t + 1). \end{cases}$$

Then, set

$$\gamma_s := \{(x, x') \in U \times \mathbf{R} ; \chi_s(x) \leq x'\}.$$

Note that γ_s is a half space when $s \leq t$ and is a closed proper cone when $s > t$. For a real number $r > 0$, writing a set

$$\begin{aligned} W_r := & \{(x, y) \in \mathbf{R}^2 ; x < r\} \cup \{(x, y) \in \mathbf{R}^2 ; y < 0\} \\ & \cup \{(x, y) \in \mathbf{R}^2 ; x < 2r, y < r, (x - 2r)^2 + (y - r)^2 > r^2\}, \end{aligned}$$

define an open set U_s in $U \times \mathbf{R}$

$$U_s := \begin{cases} \{(x, x') \in U \times \mathbf{R} ; (\text{dist}((x, x'), \gamma_s), x') \in W_{1/8s}\} & (s \geq t), \\ U \times (-\infty, t) & (s \leq t). \end{cases}$$

Then the family $(U_s)_{s \in \mathbf{R}}$ satisfies the conditions (i)–(iv) of Theorem 4.2. Indeed, for any pair $s < s'$, we have $U_s \subset U_{s'}$ and the condition (i) holds. The condition (ii) and (iv) follow by the definition of W_r . U_s has C^1 -boundary and its outer conormals belong to $-\gamma_s^\circ$, where $\circ$ means the polar cone (see [KS90, (3.7.6)]). This does not intersect $\text{SS}_k(\tilde{G})$ because of (5.2). Therefore (U_s) satisfies the condition (iii). Applying Theorem 4.2, we have the result for the morphism

$$H^j(U \times \mathbf{R}; \tilde{G}) = H^j\left(\bigcup_{s \in \mathbf{R}} U_s; \tilde{G}\right) \rightarrow H^j(U_t; \tilde{G}) = H^j(U \times]-\infty, t[; \tilde{G}),$$

and the result follows.

(b) First, we show the following lemma.

Lemma 5.2. q is proper on $\text{supp}(\text{R}i_{t*}i_t^{-1}\tilde{G})$.

Proof. Since f is proper on $\text{supp} G \cap \overline{Y_t}$, Proposition 2.1 implies that it suffices to show that $\text{supp}(\text{R}i_{t*}i_t^{-1}\tilde{G}) \subset \tilde{f}(\text{supp} G \cap \overline{Y_t})$. We have

$$\begin{aligned} \text{supp}(\text{R}i_{t*}i_t^{-1}\tilde{G}) &= \text{supp}(\text{R}\tilde{f}_* \text{R}j_{t*}j_t^{-1}G) \\ &\subset \tilde{f}(\text{supp} \text{R}j_{t*}j_t^{-1}G) \\ &= \tilde{f}\left(\overline{j_t(\text{supp} j_t^{-1}G)}\right) \\ &= \tilde{f}\left(\overline{j_t(\text{supp} G \cap Y_t)}\right) \\ &= \tilde{f}(\overline{\text{supp} G \cap Y_t}) \\ &= \tilde{f}(\text{supp} G \cap \overline{Y_t}). \end{aligned}$$

□

By the lemma, we have

$$\begin{aligned} \text{SS}_k(\text{R}q_* \text{R}i_{t*}i_t^{-1}\tilde{G}) &\subset q_\pi q_d^{-1} \text{SS}_k(\text{R}i_{t*}i_t^{-1}\tilde{G}) \\ &= \left\{ (x; \xi) \in T^*X ; \exists (x, t'; \xi, 0) \in \text{SS}_k(\text{R}i_{t*}i_t^{-1}\tilde{G}) \right\}. \end{aligned}$$

It follows from [FM07, Proposition 5.2] that

$$\begin{aligned} \mathrm{SS}_k(\mathrm{R}i_{t*}i_t^{-1}\tilde{G}) &\subset \mathrm{SS}_k(\tilde{G}) + N^*(X \times]-\infty, t[) \\ &= \mathrm{SS}_k(\tilde{G}) + T_X^*X \times \{(t; \tau) ; \tau \leq 0\}. \end{aligned}$$

Thus, for $t > t_o$, one can show

$$\begin{aligned} \mathrm{SS}_k(\mathrm{R}q_* \mathrm{R}i_{t*}i_t^{-1}\tilde{G}) &\subset \left\{ (x; \xi) \in T^*X ; \exists (x, t'; \xi, 0) \in \mathrm{SS}_k(\mathrm{R}i_{t*}i_t^{-1}\tilde{G}) \right\} \\ &\subset \left\{ (x; \xi) \in T^*X ; \exists (x, t'; \xi, 0) \in \mathrm{SS}_k(\tilde{G}) + T_X^*X \times \{(t; \tau) ; \tau \leq 0\} \right\} \\ &\subset \left\{ (x; \xi) \in T^*X ; \exists (x, t'; \xi, 0) \in \mathrm{SS}_k(\tilde{G}) \right\} \end{aligned}$$

and the result follows. $\square$

Considering the case $X = \{\mathrm{pt}\}$, one has the absolute version of the microlocal Morse lemma.

Theorem 5.3. Let $F \in \mathrm{D}^b(\mathbf{k}_M)$, and $\psi: M \rightarrow \mathbf{R}$ be a real-valued function of class C^1 , proper on $\mathrm{supp}(F)$. Let $a < b$ be real numbers, and $k \in \mathbf{Z}$. We assume $d\psi(x) \notin \mathrm{SS}_k(F)$ for $a \leq \psi(x) < b$. For $t \in \mathbf{R}$, we set $M_t := \psi^{-1}(]-\infty, t[)$. Then, the restriction morphism

$$H^j(M_b; F) \rightarrow H^j(M_a; F)$$

is an isomorphism for each $j < k$, and injective for $j = k$.

6 Proof of Lemma 4.4

In this section, we prove Lemma 4.4, which is postponed in the proof of Theorem 4.2.

Proof of Lemma 4.4 First we prove that for $F \in \mathrm{Mod}(\mathbf{k}_X)$, the natural morphism

$$(6.1) \quad \varinjlim_{t>s} \Gamma(U_t; F) \xrightarrow{\sim} \Gamma(Z_s \cup U_s; F).$$

is an isomorphism. Consider the following diagram.

$$\begin{array}{ccc}
\varinjlim_{t>s} \Gamma(U_t; F) & & \\
\downarrow \tilde{\phi} & \searrow & \\
\varinjlim_{U \supset Z_s} \Gamma(U \cup U_s; F) & \xrightarrow{\phi} & \Gamma(Z_s \cup U_s; F|_{Z_s \cup U_s}) \\
\downarrow & & \downarrow \\
\varinjlim_{U \supset Z_s} \Gamma(U; F) & \xrightarrow[\psi]{\sim} & \Gamma(Z_s; F|_{Z_s}).
\end{array}$$

Let us begin by showing that the morphism

$$\tilde{\phi}: \varinjlim_{t>s} \Gamma(U_t; F) \rightarrow \varinjlim_{U \supset Z_s} \Gamma(U \cup U_s; F)$$

is an isomorphism. It is enough to show that $(U_t)_{t>s}$ is a cofinal family of $(U \cup U_s)_{U \supset Z_s}$. Let U be an open neighborhood of Z_s . Then we shall show by contradiction that there exists a real number $t > s$ such that $U_t \subset U \cup U_s$. Let us assume that $U_t - (U \cup U_s) \neq \emptyset$ for any $t > s$. Then define the family of subsets $(A_r)_{s < r < t}$ of $\overline{U_t - U_s}$ by $A_r := U_r - (U \cup U_s)$. This $(A_r)_{s < r < t}$ satisfies the finite intersection property. Indeed, for finite real numbers $s < r_0 < \dots < r_m < t$, it follows from the assumption for contradiction that

$$\begin{aligned}
A_{r_0} \cap \dots \cap A_{r_m} &= A_{r_0} \\
&= U_{r_0} - (U \cup U_s) \\
&\neq \emptyset.
\end{aligned}$$

Therefore, since $\overline{U_t - U_s}$ is compact, we have $\bigcap_{s < r < t} \overline{A_r} \neq \emptyset$. However, we

have from the hypothesis $Z_s \subset U$ that

$$\begin{aligned}
\emptyset \neq \bigcap_{s < r < t} \overline{A_r} &= \bigcap_{s < r < t} \overline{U_r - (U \cup U_s)} \\
&= \bigcap_{s < r < t} (\overline{U_r - U_s} \cap \overline{U_r - U}) \\
&= \bigcap_{s < r < t} \overline{U_r - U_s} \cap \bigcap_{s < r < t} \overline{U_r - U} \\
&= Z_s \cap \bigcap_{s < r < t} \overline{U_r - U} \\
&= Z_s \cap \bigcap_{s < r < t} \overline{U_r} \cap \overline{X - U} \\
&= Z_s \cap \bigcap_{s < r < t} \overline{U_r} \cap (X - U) \\
&= Z_s \cap (X - U) \cap \bigcap_{s < r < t} \overline{U_r} \\
&= \emptyset.
\end{aligned}$$

This is a contradiction, and it follows that $\tilde{\phi}$ is an isomorphism.

Since $\tilde{\phi}$ is an isomorphism, it is enough to show that

$$(6.2) \quad \phi: \varinjlim_{U \supset Z_s} \Gamma(U \cup U_s; F) \rightarrow \Gamma(Z_s \cup U_s; F)$$

is an isomorphism. Since X is Hausdorff and Z_s is compact, the natural morphism

$$\psi: \varinjlim_{U \supset Z_s} \Gamma(U; F) \rightarrow \Gamma(Z_s; F)$$

is an isomorphism by Proposition 2.2. Since U_s is open, we have another isomorphism

$$\psi': \varinjlim_{V \supset U_s} \Gamma(V; F) \rightarrow \Gamma(U_s; F).$$

The injectivity of ϕ follows from the definition of a sheaf and the injectivities of ψ and ψ' .

Let us prove that ϕ is surjective. Let $\sigma \in \Gamma(Z_s \cup U_s; F)$ be a section of $F|_{Z_s \cup U_s}$. By the isomorphism ψ above, we can extend $\sigma|_{Z_s}$ to an open neighborhood U of Z_s in X . More precisely, there exists a section τ on U such that $\tau|_{Z_s} = \sigma|_{Z_s}$. On the other hand, it follows from the compatibility of restrictions of a section that $\tau|_{Z_s} = (\tau|_{(Z_s \cup U_s) \cap U})|_{Z_s}$. By the definition

of a section on the closed subset Z_s of $(Z_s \cup U_s) \cap U$, we can find an open subset W of $(Z_s \cup U_s) \cap U$ such that

$$\tau|_W = \sigma|_W.$$

By the definition of the induced topology of $(Z_s \cup U_s) \cap U$, there exists an open set $\widetilde{W}$ of X such that

$$W = ((Z_s \cup U_s) \cap U) \cap \widetilde{W} \quad \text{and} \quad \widetilde{W} \subset U.$$

Thus we have

$$(6.3) \quad U_s \cap W = U_s \cap \widetilde{W}, \quad \sigma|_{U_s \cap \widetilde{W}} = \tau|_{U_s \cap \widetilde{W}}.$$

For this $\widetilde{W}$, the pair $(U_s, \widetilde{W})$ is an open covering of $\widetilde{W} \cup U_s$. Thus by (6.3), we can glue $\sigma|_{U_s}$ and $\tau|_{\widetilde{W}}$ together and get a section $\tilde{\sigma}$ such that

$$\tilde{\sigma}|_{Z_s \cup U_s} = \sigma, \quad \tilde{\sigma}|_{\widetilde{W}} = \tau|_{\widetilde{W}}.$$

This proves the surjectivity of ϕ .

Next let $F \in \mathbf{D}^+(\mathbf{k}_X)$. Then it follows from (6.1) and Lemma 6.1 below that

$$\varinjlim_{t>s} H^j(U_t; F) \cong H^j(Z_s \cup U_s; F).$$

This completes the proof. $\square$

6.1 A Special Isomorphism of Global Sections

In this subsection, we shall show a an isomorphism of global sections which is needed in the proof of Theorem 4.2.

Let X be a topological space, and S a subset of X . we denote the derived functor of the left exact functor $F \mapsto \Gamma(S; F|_S)$ by $\mathbf{R}\Gamma(S; \cdot)$. There exists a morphism of functors

$$\mathbf{R}\Gamma(S; F) \longrightarrow \mathbf{R}\Gamma(S; F|_S),$$

which is not an isomorphism in general. Suppose the set S has the following property: for any sheaf F , the canonical morphism

$$\varinjlim_{S \subset W} \Gamma(W; F) \longrightarrow \Gamma(S; F|_S)$$

is an isomorphism, where W runs through the open neighborhoods of S .

Note that in this case, $\mathrm{R}\Gamma(S; \cdot)$ can be computed by a flabby resolution. That is, if I is a flabby resolution of F , then

$$\mathrm{R}\Gamma(S; F) \simeq \Gamma(S; I|_S).$$

The right-hand side is generally not equal to $\mathrm{R}\Gamma(S; F|_S)$, because $I|_S$ is not necessarily a complex consisting of flabby sheaves on the space S . However, in a special case, the following theorem holds.

Lemma 6.1. Let X be a Hausdorff topological space, $Z \subset X$ a compact set, and $U \subset X$ an open set, and assume $U \cap Z = \emptyset$. For $F \in \mathbf{D}^+(\mathbf{k}_X)$,

$$\mathrm{R}\Gamma(U \cup Z; F) \xrightarrow{\sim} \mathrm{R}\Gamma(U \cup Z; F|_{U \cup Z}).$$

In particular,

$$H^k(U \cup Z; F) \simeq \varinjlim_{Z \subset W} H^k(U \cup W; F),$$

where W runs through the open neighborhoods of Z .

Proof. First, if F is a sheaf, by the completely same argument in the proof of (6.2), we can show the isomorphism

$$\Gamma(U \cup Z; F|_{U \cup Z}) \simeq \varinjlim_{Z \subset W} \Gamma(U \cup W; F).$$

Let F be a sheaf and W an open neighborhood of Z . There exists an exact sequence

$$0 \rightarrow \Gamma(U \cup W; F) \rightarrow \Gamma(U; F) \oplus \Gamma(W; F) \rightarrow \Gamma(U \cap W; F) \simeq \Gamma(W; \Gamma_U F).$$

Furthermore, if F is flabby, the last morphism is surjective. Then, by letting W run through the neighborhoods of Z , we obtain an exact sequence

$$0 \rightarrow \Gamma(U \cup Z; F|_{U \cup Z}) \rightarrow \Gamma(U; F) \oplus \Gamma(Z; F|_Z) \rightarrow \Gamma(Z; (\Gamma_U F)|_Z),$$

and moreover, if F is flabby, the last morphism is surjective thanks to the exactness of the inductive limit.

Therefore, we have a distinguished triangle of derived functors

$$(6.4) \quad \mathrm{R}\Gamma(U \cup Z; F) \rightarrow \mathrm{R}\Gamma(U; F) \oplus \mathrm{R}\Gamma(Z; F) \rightarrow \mathrm{R}\Gamma(Z; \mathrm{R}\Gamma_U F) \xrightarrow{+1}.$$

Indeed, if F is flabby, so is $\Gamma_U F$, which implies

$$\mathrm{R}(\Gamma(Z; (\Gamma_U(\cdot))|_Z)) \simeq \mathrm{R}\Gamma(Z; \mathrm{R}\Gamma_U(\cdot)).$$

For simplicity, let $G = F|_{U \cup Z}$ and consider the triangle obtained by applying (6.4) assuming the whole space is $U \cup Z$. Since Z is compact and U is an open set in $U \cup Z$, we can use (6.4) to get

$$\mathrm{R}\Gamma(U \cup Z; G) \rightarrow \mathrm{R}\Gamma(U; G) \oplus \mathrm{R}\Gamma(Z; G) \rightarrow \mathrm{R}\Gamma(Z; \mathrm{R}\Gamma_U G) \xrightarrow{+1}.$$

Since Z is compact in $U \cup Z$, we have

$$\mathrm{R}\Gamma(Z; \mathrm{R}\Gamma_U G) \simeq \mathrm{R}\Gamma(Z; (\mathrm{R}\Gamma_U G)|_Z).$$

On the other hand, denoting the inclusions by $j: U \hookrightarrow U \cup Z$ and $\tilde{j}: U \hookrightarrow X$, we have

$$(\mathrm{R}\Gamma_U G)|_Z \simeq (\mathrm{R}j_*(F|_U))|_Z \simeq (\mathrm{R}\tilde{j}_*(F|_U))|_Z \simeq (\mathrm{R}\Gamma_U F)|_Z$$

and obtain

$$\mathrm{R}\Gamma(Z; \mathrm{R}\Gamma_U G) \simeq \mathrm{R}\Gamma(Z; (\mathrm{R}\Gamma_U F)|_Z) \simeq \mathrm{R}\Gamma(Z; \mathrm{R}\Gamma_U F).$$

The last isomorphism uses the fact that Z is compact in X . Similarly,

$$\mathrm{R}\Gamma(U; G) \simeq \mathrm{R}\Gamma(U; F), \quad \mathrm{R}\Gamma(Z; G) \simeq \mathrm{R}\Gamma(Z; F).$$

From the above, we obtain the triangle

$$\mathrm{R}\Gamma(U \cup Z; G) \rightarrow \mathrm{R}\Gamma(U; F) \oplus \mathrm{R}\Gamma(Z; F) \rightarrow \mathrm{R}\Gamma(Z; \mathrm{R}\Gamma_U F) \xrightarrow{+1},$$

and by comparing it with (6.4), we find that

$$\mathrm{R}\Gamma(U \cup Z; F|_{U \cup Z}) = \mathrm{R}\Gamma(U \cup Z; G) \simeq \mathrm{R}\Gamma(U \cup Z; F)$$

holds. □

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